

1 Provision & connect
POST /daq · test-driver

2 Bind node → line / device
PUT /daq/{id}/bindings

3 Compose & deploy team
POST /teams/{id}/deploy

4 Governed closed loop
dcw_control → dcw_judge

1 Provision & connect

one edge runtime per node

columns: sampling cadence · WS push · driver · line

全局 全局 mock 1号产线

5000ms 1000ms

1000ms 全局 modbus-tcp 1号产线

1000ms

Each device becomes one **node**: cadence, **driver** and line come from its export, not code---modbus-tcp and replayed mock nodes share one loop.

live line 1: 9 nodes, 4 protocol families, 1 s and 5 s cadences.

3 Compose & deploy the agent team

one lead, harness-backed workers

AML 影子建模团队

内置 system 部署

lead AML 首席数据科学家 omp
worker AML 数据工程师 omp
worker AML 训练工程师 omp

A team is a reusable **role / harness** assignment---one lead plus workers, each backed by one of 14 interchangeable engines---deployed into a channel.

ready lead + workers on harness omp; a mock team enables replay.

2 Bind the node to the plant

device + recipe window travel with the node

设定下发	工艺量程	保写周期	绑定设备
150~200 下发	150 ~ 200 °C	手动 · 默认 5s	dev-mtgnirdo-v70w4
0.6~1.2 下发	0.6 ~ 1.2 MPa	手动 · 默认 5s	dev-mtgnirdo-v70w4
150~200 下发	150 ~ 200 °C	手动 · 默认 5s	—

Binding gives a control node a **device twin** and a **recipe window**; both are node properties, so a deployment is re-specified by configuration.

bound 150-200 °C window, 5 s keep-alive, device dev-mtgnirdo-v70w4.

4 Governed closed loop

write → observe → judge → journal

优化记录

judged-keep 183 → 178 keep(agent)
superseded 182 → 183 uncertain(system)

A write through dcw_control opens an **optimization record**; dcw_judge resolves it to keep or rollback with a reason, beside the ledger's setpoint history.

judged kept: 183 → 178 keep(agent); the superseded record carries the system's own verdict.